DO NOW — RECALL

Quicksort: pivot and partition

Name: _____ Class: _____ Date: _____

1. In bubble sort, which value reaches its final position at the end of pass one?

2. What is a base case?

3. Which element is a[3] in [6,2,9,4,5]?

4. What is the result of 4 <= 6 and why?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The largest value in the unsorted section.

2. Model answer

A condition that stops recursive calls.

3. Model answer

4; indexes begin at 0.

4. Model answer

True, because 4 is less than or equal to 6.

Vocabulary — know these terms

pivot — Element selected to split a quicksort partition.

partition — Rearrange values so smaller/equal values are on one side of the pivot.

Lomuto — A partition approach scanning left to right and placing the pivot last.

index — Zero-based position within the array.

divide and conquer — Solve smaller subproblems and combine results.

NEW LEARNING

Read, annotate and explain

Pick a pivot

Use the rightmost element, 6. A left boundary i tracks where the next small value belongs.

Scan and move

Only values less than or equal to 6 move left. We move 2, then 4, without sorting those two values.

Place the pivot

Swap 6 into index 2; elements to the left are <=6, and those to the right are >6. Each side still needs sorting.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Partition [6,3,8,2,5] around final-element pivot 5.

Predict the next state

6 is larger than 5; 3 is smaller and swaps into the left boundary.

Reveal and explain

8 is larger; 2 swaps to index 1, giving [3,2,8,6,5].

Reveal and explain

Swap pivot 5 with position 2: [3,2,5,6,8]. The two sides are not yet necessarily sorted.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Partition [4,1,3] using the final value as pivot, with Lomuto partition. Give the result and pivot index.

Question 2

Partition [9,2,5,1,6] using the final pivot. Show the array after moving each qualifying value and after placing the pivot.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why choosing the last element as pivot can lead to $O(n^2)$ work on already sorted data, and suggest an improvement.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A quicksort partition automatically fully sorts the two sides of the pivot.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.